

Jai Bardhan

RESEARCHER · CIIRC, CZECH TECHNICAL UNIVERSITY IN PRAGUE

Advised by Josef Sivic and Vladimir Petrik

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Research Interests

World models for robotics: 3D-consistent action-conditioned video models, world action models, RL post-training of world models, and evaluating and improving policies with them. Broadly: generative and self-supervised learning.

Education

International Institute of Information Technology, Hyderabad

Hyderabad, India

B.TECH. COMPUTER SCIENCE & M.S. BY RESEARCH, COMPUTATIONAL NATURAL SCIENCES

Aug 2018 – Jul 2023

- CGPA **9.58/10**; **Gold Medal** (highest CGPA, Computational Natural Sciences, 2018 batch); Dean's List 2018–2022
- M.S. thesis: *Revisiting Deep Learning for Particle Physics* (advisor: Subhadip Mitra); **Ritesh Tiwari Outstanding MS Thesis Award**

Publications

ROBOT LEARNING & WORLD MODELS

- [P1] **DepthWorld: 3D World Model for Robot Manipulation**
Jai Bardhan, Josef Sivic, Vladimir Petrik
Conference on Robot Learning (CoRL) 2026 [arXiv] [project] [code] [DROID-3D data: 71k recalibrated episodes]
- [P2] **Persistent Robot World Models: Stabilizing Multi-Step Rollouts via Reinforcement Learning**
Jai Bardhan, Patrik Drozdík, Josef Sivic, Vladimir Petrik
European Conference on Computer Vision (ECCV) 2026 • **Oral & Outstanding Paper Award, UniWorld @ ECCV** [arXiv] [project] [code]
- [P3] **REALM: A Real-to-Sim Validated Benchmark for Generalization in Robotic Manipulation**
Martin Sedlacek, Pavlo Yefanov, Georgy Ponimatin, **Jai Bardhan**, Simon Pilc, Mederic Fourmy, Evangelos Kazakos, Cees G. M. Snoek, Josef Sivic, Vladimir Petrik
IEEE Robotics and Automation Letters (RA-L) 2026 [arXiv]

OTHER PUBLICATIONS

- [P4] **Curve Skeletonization in Continuous Domain**
Jai Bardhan, Ramya Hebbalaguppe, Aravind Udupa
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026
- [P5] **Refine and Align: Confidence Calibration through Multi-Agent Interaction in VQA**
Ayush Pandey, **Jai Bardhan**, Ishita Jain, Ramya Hebbalaguppe, Rohan Raju Dhanakshirur, Lovekesh Vig
AAAI Conference on Artificial Intelligence (AAAI) 2026, Special Track on AI Alignment
- [P6] **Tagging fully hadronic exotic decays of the vectorlike B quark using a graph neural network**
Jai Bardhan, Tanumoy Mandal, Subhadip Mitra, Cyrin Neeraj, Mihir Rawat
Physical Review D 112, 015015 (2025) (alphabetical author order) [paper]
- [P7] **HEP-JEPA: A foundation model for collider physics using joint embedding predictive architecture**
Jai Bardhan, Radhikesh Agrawal, Abhiram Tilak, Cyrin Neeraj, Subhadip Mitra
ICLR 2025 Workshops on World Models and on Agentic AI for Scientific Discovery [arXiv]
- [P8] **Constructing sensible baselines for Integrated Gradients**
Jai Bardhan, Cyrin Neeraj, Mihir Rawat, Subhadip Mitra
AI to Accelerate Science and Engineering (AI2ASE) Workshop @ AAAI 2025
- [P9] **Loss function to optimise signal significance in particle physics**
Jai Bardhan, Cyrin Neeraj, Subhadip Mitra, Tanumoy Mandal
Machine Learning and the Physical Sciences Workshop @ NeurIPS 2024
- [P10] **ReMOVE: A Reference-free Metric for Object Erasure**
Aditya Chandrasekar, Goirik Chakrabarty, **Jai Bardhan**, Ramya Hebbalaguppe, Prathosh AP
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR-W) 2024
- [P11] **Machine learning-enhanced search for a vectorlike singlet B quark decaying to a singlet scalar or pseudoscalar**
Jai Bardhan, Tanumoy Mandal, Subhadip Mitra, Cyrin Neeraj
Physical Review D 107, 115001 (2023) [paper]

REVIEW ARTICLES

- [R1] **Unsupervised and lightly supervised learning in particle physics**
Jai Bardhan, Tanumoy Mandal, Subhadip Mitra, Cyrin Neeraj, Monalisa Patra
The European Physical Journal Special Topics (2024)
- [R2] **Deep reinforcement learning in chemistry: A review**
Bhuvanesh Sridharan, Animesh Sinha, **Jai Bardhan**, Rohit Modee, Masahiro Ehara, U. Deva Priyakumar
Journal of Computational Chemistry (2024)

UNDER REVIEW & IN PREPARATION

- [S1] **KEEP: Knowledge-Preserving Expert Distillation for Embodied Policies**
Georgy Ponimatin, William Balucha, Lenka Šimová, Patrik Drozdík, Martin Sedlacek, **Jai Bardhan**, Evangelos Kazakos, Josef Sivic, Vladimir Petrik
Under review at ICLR 2027
- [S2] **Confidence-Guided Query Refinement for Search-Augmented Reasoning**
Ayush Pandey, **Jai Bardhan**, Ramya Hebbalaguppe
Under review at ICLR 2027
- [S3] **HEP-JEPA2: Grounded, Multi-Depth Latent Prediction for Self-Supervised Jet Representations**
Jai Bardhan*, Cyrin Neeraj*, Abhiram Tilak*, Subhadip Mitra
In preparation * equal contribution

Research Experience

Researcher

Prague, Czechia

IMPACT GROUP, CIIRC, CZECH TECHNICAL UNIVERSITY IN PRAGUE

Sep 2025 – Present

- **PersistWorld [P2]**: designed and implemented an RL post-training method that keeps video world models stable over long autoregressive rollouts. **14% lower LPIPS** than Ctrl-World; on a real-robot 4-block picking task, π_0 trained on its rollouts reaches **0.67 task progression vs. 0.56** with Ctrl-World's.
- **DepthWorld [P1] & DROID-3D**: built a 3D world model that predicts metric depth alongside RGB (**+1.48 dB PSNR** on external views at equal training budget), and the calibration pipeline behind its training data, recalibrating **71k DROID episodes** from 28 robots to 4.9 mm median extrinsic error against ChArUco ground truth (per-scene baseline: 72 mm). Released code and checkpoints; data available on request.
- **KEEP [S1]**: helped design the core method, the distillation loss and the experiments.
- **REALM [P3]**: helped port the benchmark to a newer Isaac Sim version and contributed to its evaluation metrics.

Predoctoral Research Fellow

Delhi, India

TCS RESEARCH, DEEP LEARNING AND AI LAB

Jun 2023 – Jun 2025

- **Curve Skeletonization [P4]** (WACV 2026): project lead; 3D shape skeletonization via discrete differential geometry.
- **Refine & Align [P5]** (AAAI 2026): designed and derived the loss; helped design the experiments.
- **ReMOVE [P10]** (CVPR-W 2024): helped design the reference-free object-erasure metric and its experiments.
- **Query Refinement [S2]**: helped design the experiments.
- Explored interpretability of personalized diffusion models via unsupervised concept discovery.

Research Assistant

Hyderabad, India

CENTER FOR COMPUTATIONAL NATURAL SCIENCES AND BIOINFORMATICS, IIIT HYDERABAD

May 2020 – May 2023

- **New-physics searches**: led an ML-enhanced search for vectorlike B quarks (Phys. Rev. D 2023) [P11]; advised and contributed experiments to GNN tagging of their decays (Phys. Rev. D 2025) [P6].
- **ML methods for physics**: led HEP-JEPA, a self-supervised foundation model for collider data [P7], baselines for Integrated Gradients [P8], and a loss that directly optimizes signal significance [P9].
- **Mentoring**: co-advised two master's students with Subhadip Mitra (Abhiram Tilak, Mihir Rawat), leading to [P6–P8] and [S3].
- Led a review of unsupervised learning in particle physics [R1]. M.S. thesis on deep learning for particle physics.

GSoC Contributor

Remote

MACHINE LEARNING FOR SCIENCE (ML4SCI), GOOGLE SUMMER OF CODE

Aug 2022 – Oct 2022

- Built a GraphGPS-based graph neural network for end-to-end jet mass regression on LHC particle-shower data. [blog]

Research Intern

Tokyo, Japan

SONY GROUP CORPORATION, CREATIVE AI LAB

May 2022 – Jul 2022

- Single-image 3D clothed-human reconstruction: analyzed failure modes of ICON (then SOTA) and designed architectural fixes.

Research Student

Hyderabad, India

MACHINE LEARNING LAB, IIIT HYDERABAD

Oct 2021 – Feb 2022

- Domain adaptation for point clouds: adapted SimCLR, DINO and BYOL to 3D data, combined with feature alignment.

Awards & Service

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| 2026 | Outstanding Paper Award & Oral , UniWorld Workshop @ ECCV 2026, for PersistWorld [P2] | Malmö, Sweden |
| 2026 | Gold Reviewer , International Conference on Machine Learning (ICML 2026) | |
| 2024–26 | Reviewer , CoRL, ECCV, CVPR, SIGGRAPH Asia, BMVC, NeurIPS, ICML, AAAI, TMLR | |